

JOSEPH SPENCER

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EDUCATION

University of Utah: Salt Lake City, UT

09/2025- present

Doctor of Philosophy, Metallurgical Engineering

Advisor: Dr. Zak Fang

- Improved laser powder bed fusion manufacturing method for titanium parts
- Restored an ExOne binder jet printer to operation to measure printed and sintered parts
- Designed sintering deformation models for ANSYS software
- Use ANSYS optiSlang to determine unknown properties of metals

University of Utah: Salt Lake City, UT

09/2023-04/2025

Master of Science, Mechanical Engineering, Data Science Focus

Dean's List

- Extracted mechanical properties of materials from limited experimental data
- Developed machine learning models using deep neural networks
- Created models of nanoindentation using the finite element method

Brigham Young University: Provo, UT

09/2018-04/2023

Bachelor of Science, Mechanical Engineering, Aerospace Emphasis

Presidential Scholar

- Senior Design: X-ray tube heat removal system for Varex Imaging

RESEARCH EXPERIENCE

Research and development intern

06/2025-08/2025

[IperionX Critical Minerals](#) – Dr. Matt Dunstan

- Evaluated various mixing conditions using ANSYS Fluent to assist product development.
- Developed standard protocol for modeling fluid mixing in ANSYS Fluent.

Research fellow

06/2024-08/2024

Universities Space Research Association – [Center for Space Nuclear Research](#) – Dr. Robert O'Brien

- Designed a liquid droplet radiator for cooling a nuclear reactor in outer space.

Research assistant

01/2023-05/2023

Brigham Young University – [FLOW Lab](#) – Dr. Andrew Ning

- Use various Julia software packages to model and optimize the behavior of rotors and propellers

Research assistant

05/2021-06/2022

Brigham Young University – [CREATE Lab](#) – Dr. Nathan Crane

- Performed experiments to improve the quality of parts manufactured by binder jetting
- Operated an ExOne Innovent+ binder jetting 3D printer
- Designed and manufactured a protective covering for a custom printer used in the lab

Research assistant

06/2019-07/2019

Brigham Young University – [Ess Lab](#) – Dr. Dan Ess

- Simulated organometallic chemical reaction dynamics using density functional theory
- Left after suffering a severe injury in a bicycle accident

TEACHING EXPERIENCE

Teaching assistant

09/2023-04/2025

The University of Utah

- Mechanics of Materials (ME EN 3310) – Dr. Ken Monson (Fall 2023), Dr. Owen Kingstedt (Winter 2025)
- Applied Finite Elements (ME EN 6510) – Dr. Jake Hochhalter (Fall 2024)
- Strengths of Materials Lab (ME EN 3315) – Dr. Ken Monson (Fall 2023), Dr. Owen Kingstedt (Winter 2024, Winter 2025)

- Created in-class quizzes to measure student understanding of lectures.
- Created a lab to teach finite element analysis of electromagnetic systems.
- Taught students how to measure the mechanical properties of a variety of materials and ensure their safety
- Graded homework and lab submissions

Grading assistant

01/2023-04/2023 Brigham Young University

- Composite Manufacturing (ME EN 456) – Dr. David Fullwood (Winter 2023)
- Engineering Measurements (ME EN 362) – Dr. Jeff Hill, Dr. Brady Davies (Winter 2023)
- Graded engineering measurements lab submissions
- Graded composite manufacturing homework assignments and exams

INDUSTRY EXPERIENCE

[Blue Raven Solar](#)

05/2022-08/2022 Design Engineer, Proposals Engineer

- Used 3D computer-aided design software to create both initial and final designs of residential solar panel systems

CONFERENCE PROCEEDING

1. Lawrence, J., Pera Vega, H. A., Stegman, B., Roberts, C., **Spencer, J.**, James, C., Christensen, M., Crane, N. (2022). [Construction of Open-Source Laboratory- Scale Binder Jetting System for High-Speed Synchrotron X-Ray Imaging](#). Solid Freeform Fabrication 2022: Proceedings of the 33rd Annual International Solid Freeform Fabrication Symposium – An Additive Manufacturing Conference, 1508-1518.

CONFERENCE POSTER PRESENTATION

1. **Spencer, J.**, Inckley, C., Crane, N. (2021) Measuring the Effects of Heat Lamp Exposure and Bottom Bleed Reduction on Binder-jetted Parts, Utah Conference on Undergraduate Research, Utah Tech University.

QUALIFICATIONS

Programming Languages:

- C++ (Advanced)
- Python (Advanced)
- Julia (Intermediate)
- MATLAB (Intermediate)
- R (Intermediate)

Computer Programs:

- $\text{\LaTeX}$ (Advanced)
- ANSYS (Advanced)
- MOOSE (Intermediate)
- Coreform Cubit (Intermediate)
- SolidWorks (Intermediate)

Language:

- English: Native
- Spanish: Fluent

PROFESSIONAL MEMBERSHIP

- American Society of Mechanical Engineers *02/2024-Present*
- University of Utah Scientific Computing Student Club (SC)² *09/2023-04/2025*
- Delta Alpha Pi academic honor Society *09/2020-04/2023*

SCHOLARSHIPS

- Dayna Lynn Ahern scholarship *09/2021, 01/2022, 09/2024, 06/2026*
 - National Merit scholarship *09/2018*
 - Thomas S Monson presidential scholarship *09/2018*
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